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Effectiveness of the tropical plants *Rhynchospora corymbosa* and *Coix lacryma-jobi* in vertical flow constructed wetlands for municipal primary sewage effluent treatment

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ABSTRACT

The performance of two tropical plants, *Rhynchospora corymbosa* L. (RC) and *Coix lacryma-jobi*, L (CL) in treatment of primary sewage effluent in lab-scale vertical-flow constructed wetlands (VFCW) along with no plant control wetland was investigated. A batch-flow VFCWs were operated under batch fill and drain hydraulic loading system with hydraulic retention times (HRT) of 0.5, 1, and 2 days and fill rate of 8 L/day. Removal of solids, organics, nutrients, and pathogens were monitored. The volumetric contaminant removal rates were best described by 1st order kinetics except for ammonia and phosphate, which was best described by Stover-Kincannon kinetics. Influent TSS, PO_4^{3-}, COD, BOD_5, and total coliform concentration were low but high in NH_4^+ concentration. CL was better in nutrient removal as HRT increases compared to RC. RC was more efficient at TSS, turbidity, and organics removal. Pathogen removal was independent of plant type but HRT. Solids and organic removal were lower in CL planted CWs due to preferential flow paths created by their bulky root. CL planted CWs removed more nutrients followed by RC planted CWs and then no-plant control CWs. The results of these tests demonstrate that both CL and RC are suitable for the treatment of municipal wastewater in VFCW system.

KEYWORDS

Coix lacryma-jobi L; coliform removal; constructed wetland; hydraulic retention time; primary sewage; *Rhynchospora corymbose* L

Introduction

As industrialization continues to grow, high-quality freshwater resources are becoming more difficult to find and protect. An assessment of surface water bodies conducted in the United States of America revealed that 40–50% are impaired or threatened (USEPA 2005). These water bodies are impacted by both natural and anthropogenic means, including untreated wastewater discharge. Water pollution, particularly through wastewater discharge, is seen as a threat to human and environmental health because of the transmission of bacterial and viral waterborne diseases and the eutrophication of water bodies leading to potentially toxic algal blooms and low dissolved oxygen concentrations (Cooper *et al.* 2020). The impact of sewage-contaminated surface waters is greatest in developing countries where waterborne diseases remain a major public health threat (WHO and UNICEF 2017).

In Sub-Saharan Africa and Southern Asia, less than half of the population have access to proper sanitation, with millions of people dying yearly, and about 3,900 children dying every day, because of transmitted disease through untreated water disposal and human excreta contamination of surface water (Zhang *et al.* 2015). The centralized collection, treatment, and discharge of sewage are often considered to be the optimal solution in terms of treatment and pollution control. This method is commonly deployed in developed counties and has effectively solved the problems of sanitation in these countries (Gallego-Schmid and Tarpani 2019). However, centralized primary sewage effluent treatment has limitations in developing countries, including high capital cost, improper operation, and dependent on treatment technologies that are unaffordable in areas with low population densities and dispersed households (Rashid and Liu 2020). Additionally, these high-technology wastewater treatment systems may not be feasible in many developing countries due to their requirements for a regular power supply, the availability of replacement parts, and access to trained personnel for operation and maintenance (Gallego-Schmid and Tarpani 2019). Thus, simpler designs are required for the practical and successful treatment of wastewater in the developing regions of the world. The construction of artificial wetlands for wastewater treatment is now a widely approved and rapidly growing alternative to conventional treatment (Ewemoje *et al.* 2015; Abdelhakeem *et al.* 2016).

Constructed wetlands (CW) are engineered systems that have undergone design and construction to make use of natural processes which include vegetation, soils or gravels of

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various diameters, and microbial assemblages to help in the treatment of wastewater. They are designed to function like natural wetlands but with a much greater degree of control (Vymazal 2011a). Constructed wetlands are effective in the removal of organic carbon, pathogens, and nutrients in wastewater before final release into a body of water (Jiang *et al.* 2016; Lamori *et al.* 2019). The majority of CW studies use plants found native to Europe and North America that can withstand cold winters. The plants that have most commonly been investigated for treatment purposes include cattails (*Typha* spp), bulrushes (*Scirpus lacustris*), and reeds (*Phragmites australis*) (Zhang *et al.* 2015; Sandoval *et al.* 2019; Tang *et al.* 2019; Lei *et al.* 2023). While the majority of studies have used plants native to Europe and North America, the region of the world that would most directly benefit from CW are generally found in tropical and subtropical climates. Thus, there is a need to identify indigenous tropical plants that would be suitable for use in CW in more tropical climates (Bohórquez *et al.* 2017; Mustapha *et al.* 2018; Varma *et al.* 2021). According to Lei *et al.* (2023) Phytoremediation is often applied by means of constructed wetlands to treat wastewater.

Rhynchospora corymbosa L. Britton is a member of the sedge family found along swamps, streams, and lowlands but common in the tropical environment. Adekanmbi *et al.* (2009) counted over 55 plants stands in an area of 900 m^2 at three different locations in the mangrove environment in southwestern Nigeria. The plant is found in abundance in the study area. The plant without stolon is a perennial, emergent aquatic macrophytes. *Rhynchospora corymbosa* L. Britton is a fast-growing rhizomatic plant that is best propagated vegetatively (Taha *et al.* 2015; Ewemoje *et al.* 2019). *Rhynchospora corymbosa* L. Britton has been reported to be efficient in the treatment of primary sewage lagoon effluent in VFCW (Ewemoje and Sangodoyin 2018).

Coix lacryma-jobi L. commonly known as Job's Tear is a tall broad-leaf tropical plant of South Asia origin (Jampeetong *et al.* 2013; Patel and Parikh 2020). *Coix lacryma-jobi* L. has been reported to be efficient in nutrient removal from polluted effluent (Xu and Li 2007). The plant is a perennial grass with a strong root system (Chen *et al.* 2015; Nghiem *et al.* 2016) which makes it adaptable in CW system.

The wild species is an herbaceous plant found along waterways in the study area. The plant is easily cultivated from seeds. *Coix lacryma-jobi* L. has been used in a VFCW to remove Chromium in Cr (vi) bearing domestic sewage (Li *et al.* 2017). Ewemoje *et al.* (2015) investigated the potential of *C. lacryma-jobi* L. in the treatment of anaerobic lagoon in an HFCW. The result showed that the CW performed well in the removal of PO4, Ammonia, Ammonium, COD, TDS, DO, and TSS at a high HRT of 7 days. Most research involving *C. lacryma-jobi* L. focused more on nutrient removal, not pathogens. There are limited studies on the pollutant removal potential of these two studied plants in VFCW for sewage treatment. The targeted pollutants have been organics and inorganics but not on pathogens. This study compared and contrasted the performance of two aquatic emergent tropical plant *Hynchospora corymbosa* L. Britton and *C. lacryma-jobi* L.) to each other and a no plant control, under a variety of hydraulic retention times (HRTs) in the removal of organics, nutrients, solids and pathogens from primary sewage effluent in a VFCW.

Materials and methods

Constructed wetland plant preparation

Mature seeds of *C. lacryma-jobi* L. (CL) and *R. corymbosa* L. Britton (RC) plants were collected from their natural habitats far from pollution sources in western Nigeria. Seeds were treated with 50–60% H_2SO_4 for 5 min and rinsed in several changes of tap water with baking soda to neutralize it. The acid-scarified seeds were planted into nursery pot directly using a soilless media of organic mix of 55–65% Canadian sphagnum peat moss, and perlite which contains mixtures of pumice, dolomite and limestone in a greenhouse. The pots were irrigated and incubated on a heating mat set to 37.8 °C. After 8 days, both CL and RC seeds sprouted. The seedlings were transferred into the pots and NPK 20/20/20 fertilizer was applied to it weekly. Six weeks after germination, the CL and RC sprouts were transplanted equidistantly at a 5 cm interval in the constructed wetland. The plants were fed with primary effluent wastewater for four months before experimentation to create a dense bed of vegetation and establish their associated microbiome. During this time, a constructed wetland without plants was also fed with primary effluent wastewater to establish its microbiome also.

Description of pilot-scale constructed wetland

This research was conducted at the lab-scale at Oregon State University, Corvallis, Oregon, United States of America (51.5°N, 12.9°E) from September 2014 to June 2015. The wetland system was housed in a greenhouse at the Oregon State University Agricultural Research Station. Temperature and humidity control in the greenhouse permitted the simulation of climatic conditions of the plants' native environment. The greenhouse room temperature ranged from 24.7 to 29.1 °C in the summer and 22.0–24.7 °C in the fall. The constructed wetland system consisted of three units with each unit containing two cells connected in series (Figure 1). Two of the constructed wetlands were vegetated with six plants of either RC or CL, while the non-planted (NP) unit served as a control. The cells were made of cylindrical plastic tanks; each tank was 0.69 m in height, and 0.33 m in diameter with a surface area of 0.085 m^2 and a volume of 0.059 m^3. Each tank has a drain at the bottom with a 12.7 mm diameter pipe and valve control that drained into the second cell. Each cell was packed with coarse 40 mm diameter river pebbles and 7–10 mm diameter pea gravel. The total depth of the packed bed was 0.6 m, with coarse gravel at 20 cm and pea gravel at 40 cm depth in the reactor. The average substrate porosity was 0.4. Each constructed wetland received primary treated domestic

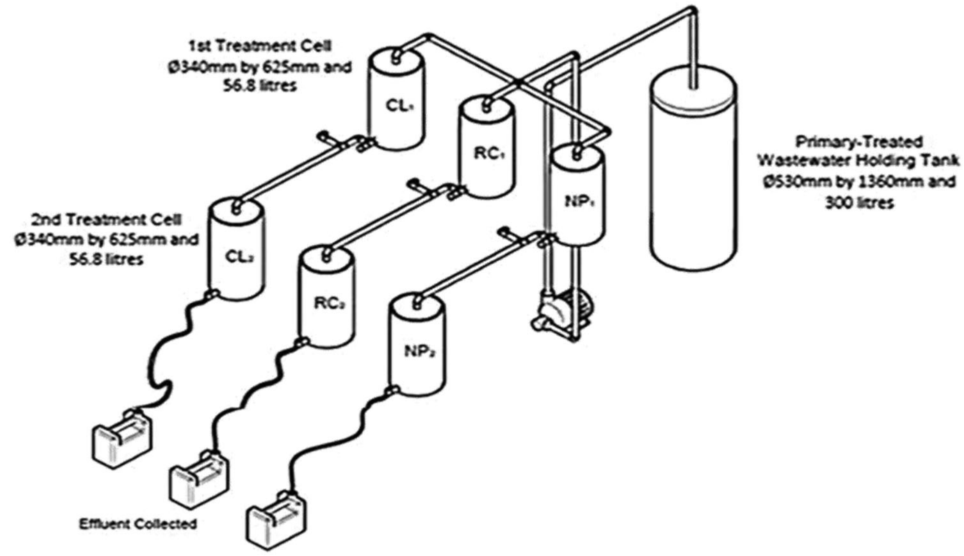

Figure 1. Constructed wetlands two sets of constructed wetlands in series were constructed containing either *C. lacryma-jobi* (CL_1 and CL_2), *R. corymbosa* (RC_1 and RC_2), or no plants (NP_1 and NP_2). All were fed the same primary treated domestic wastewater.

wastewater (from the Corvallis, OR wastewater treatment plant) in a sequential batch fill and drain process at a hydraulic filling rate of 8 L/day, and HRTs of 0.5, 1, and 2 days were tested.

Sample collection and analysis

Three sampling points were selected within each constructed wetland; (1) influent (same composition for RC_1, CL_1, and NP_1) (2) effluents from RC_1, CL_1, and NP_1; (3) effluents from RC_2, CL_2, and NP_2. Samples were taken twice a day for 0.5 days HRT, once a day for the 1-day HRT, and every other day for the 2 days HRT. All wastewater samples were analyzed using U.S. Environmental Protection Agency (USEPA) approved methods. Immediately after collection, the DO, pH, and temperature were determined using a Hach sensION156 DO meter, following the manufacturer's instructions (HACH, Loveland, CO, USA). The samples were transported immediately on ice to the laboratory for analyses of the 5-day biological oxygen demand (BOD_5), chemical oxygen demand (COD), ammonia-nitrogen (NH_4^+-N), total phosphorus (TP), nitrate-nitrogen (NO_3^--N), total suspended solids (TSS), turbidity, total coliform bacteria, and *Escherichia coli*. The BOD_5, TSS, and turbidity were determined using standard methods (APHA *et al*. 2012) TNT plus vial test were used to colorimetrically quantify the COD, NH_4^+-N, NO_3^--N, and TP per manufacturer's instructions (HACH, Loveland, CO, USA). Total coliform bacteria and Escherichia coli were quantified *via* the most probable number (MPN) method using the IDEXX Quanti-Tray/2000 with Colilert, per manufacturer's instructions (IDEXX, Westbrook, ME, USA).

Constructed wetland performance analyses

The effect of plant type and HRT on constructed wetland performance was quantified based on percentage removal, volumetric mass removal rate (r), 1st order volumetric rate constants (k), and Stove-Kincannon maximum utilization rate (U_{MAX}) and saturation (KB) constants. The percent removal (removal efficiency) was calculated as follows.

$$\text{Removal efficiency } (\%) = \frac{C_{in} - C_{out}}{C_{in}} \times 100 \qquad (1)$$

Where: C_{in} = influent concentration (g/m^3, NTUs or MPN/100 mL); C_{out} = effluent concentration (g/m^3, NTUs or MPN/100 mL); The volumetric mass removal rate was calculated as follows:

$$r = \frac{C_{in} - C_{out}}{HRT} \qquad (2)$$

Where: r = volumetric mass removal rate (g/m^3-d, NTU/d, or MPN/100 mL-d); HRT = hydraulic retention time (d)

The 1st order volumetric rate constant was calculated utilizing the 1st order plug flow reactor decay model set to zero (Abdelhakeem *et al.* 2016).

$$\ln\left(\frac{C_{out}}{C_{in}}\right) + K * HRT = 0 \qquad (3)$$

k = 1st order volumetric rate constant (d^{-1})

The generalized reduced gradient (GRG) non-linear least squares method was used to solve for k, as previously described in Lasdon *et al.* (1978) and Smith *et al.* (1998). The GRG non-linear least squared method was also used to solve for the Stover-Kincannon maximum utilization rate and saturation coefficients where the Stover-Kincannon model was set to zero (Gholizadeh *et al.* 2015).

$$\frac{U_{MAX*}C_{in}}{(K_b + C_{in})} - \frac{C_{in} - C_{out}}{HRT} = 0 \qquad (4)$$

Where: U_{MAX} = maximum substrate utilization rate (g/m^3-d); K_b = substrate saturation coefficient (g/m^3-d)

Model performance was quantified using three statistical parameters including the coefficient of determination (R^2), the relative root mean square error (RRMSE), and the model efficiency (ME) (Saeed and Sun 2011).

$$R^2 = \frac{\sum_{i=1}^{N}(X_i - \overline{X})(Y_i - \overline{Y})^2}{\sum_{i=1}^{N}(X_i - \overline{X}) \sum_{i=1}^{N}(Y_i - \overline{Y})^2} \tag{5}$$

$$\text{RRMSE} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2}}{\overline{Y}} \tag{6}$$

$$\text{ME} = \frac{\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{N}(Y_i - Y_i)^2} \tag{7}$$

Where: R^2 = coefficient of determination (0–1); RRMSE = relative root mean square error (0–∞); ME = model efficiency (−∞–1); X_i = Observed volumetric loading rate (g/m^3-d, NTU/d or MPN/100 mL-d); X = Mean observed volumetric loading rate (g/m^3-d, NTU/d or MPN/100 m; Y_i = Observed volumetric removal rate (g/m^3-d, NTU/d or MPN/100 mL-d); $\overline{Y}$ = Mean observed volumetric removal rate (g/m^3-d, NTU/d or MPN/100 mL-d); $\hat{Y}$ = Predicted volumetric 250 removal rate (g/m^3-d, NTU/d or MPN/100 mL-d)

Results and discussion

Wastewater influent characterization

For the most part, all influent wastewater parameters were statistically similar ($p > 0.05$) for each of the three HRT experiments (Table 1). However, there were a few notable exceptions. The turbidity of the 1-day HRT experiment was significantly lower ($p < 0.05$) at 50 ± 5 NTUs, compared to the 67 ± 13 NTUs and 60 ± 7 NTUs of the 0.5- and 2-day HRT experiments, respectively. Additionally, the average *E. coli* influent concentration of the 2-day HRT experiment was statistically lower ($p < 0.05$) than that of the 0.5 and 1 day HRT experiments at 8.4 × 104 ± 2.9 × 104 MPNs/100 mL, 7.9 × 105 ± 3.0 × 105 MPNs/100 mL and 7.3 × 105 ± 2.1 × 105 MPNs/100 mL, respectively. The influent wastewater is considered to be "low" strength in terms of TSS, PO_4^{3-}, COD, BOD$_5$, and total coliform and "high" strength in terms of NH_4^+. This agrees with the findings of Tchobanoglous *et al.* (2003).

Solids removal

The RC$_1$ and NP$_1$ demonstrated significantly similar TSS reductions ($p > 0.05$) ranging from 84 ± 7 to 93 ± 2% (Tables 2–4) after the first pass, under all three HRT conditions. However, CL$_1$ Performed significantly worse ($p < 0.05$) than both RC$_1$ and NP$_1$ under the three HRT conditions, by achieving only 71 ± 5 to 82 ± 3% reductions in TSS. The TSS concentrations significantly decreased ($p < 0.05$) under all conditions tested, with TSS removal ranging from 85 ± 4 to 95 ± 3% after passing through the second constructed wetland. Generally, after the second pass, there was no significant difference ($p > 0.05$) between the three plant conditions under each HRT condition, with one exception. Under a 1-day HRT, the CL$_2$ effluent contained 7.9 ± 1.7 mg/L TSS, which was significantly higher ($p < 0.05$) than either the RC$_2$ (2.5 ± 0.7 mg/L) or NP$_2$ (2.4 ± 0.6) effluents. Similar to the TSS results, after the first pass, under all three HRT conditions, the RC$_1$ and NP$_1$ constructed wetlands demonstrated significantly similar turbidity reduction ($p > 0.05$) ranging from 72 ± 6 to 93 ± 1%.

Also, the CL$_1$ constructed wetlands performed significantly worse ($p < 0.05$) than both the RC$_1$ and NP$_1$ constructed wetlands under all three HRT conditions, by achieving only 53 ± 8 to 81 ± 7% reductions in turbidity. Significant decreases ($p < 0.05$) in turbidity were observed under all conditions tests after passing through the second constructed wetland, ranging from 90 ± 2 to 98 ± 1% removal. However, unlike the TSS results, the turbidity in the CL$_2$ effluent was still significantly higher ($p < 0.05$) than the turbidity of the RC$_2$ and NP$_2$ constructed wetlands under all HRT conditions.

The TSS removals observed in this study fall within the 62–98% (84.9% average) removals that have been previously reported for tropical/sub-tropical vertical flow constructed wetlands treating municipal wastewater (Zhang *et al.* 2015) TSS removal appeared to be independent of HRT while turbidity significantly improved ($p < 0.05$) from 0.5-day HRTs to 1-day HRTs but not from 1-day HRTs to the 2-day HRT.

These suggest that the majority of the TSS is relatively settle able and can be removed quickly; however, a significant fraction of the turbidity is relatively non-settable and required longer HRTs to settle out of solution. The volumetric removal rates of TSS and turbidity followed 1st order kinetics with increased volumetric loading rates leading to increased volumetric removal rates as shown in Figure 2.

Both RC and NP constructed wetlands demonstrated similar 1st order kinetic rate constants for TSS and turbidity ranging from 4.2 to 4.5 and 2.7 to 3.0 d^{-1}, respectively (Tables 5 and 7). The TSS and turbidity 1st order rate constants were substantially lower in the CL constructed wetlands at 2.2 and 1.7 d^{-1}, respectively (Table 6). The lower

Table 1. Primary treated wastewater composition during the 0.5, 1, and 2 d HRT experiments.

	0.5 day HRT	1 day HRT	2 day HRT
TSS (mg/L)	65.1 ± 11.2	62.6 ± 10.4	59.3 ± 10.3
PH	67.1 ± 12.8	49.8 ± 4.6	60.3 ± 7.3
PO_4^{3-} (mg/L-P)	7.33 ± 0.07	7.19 ± 0.05	7.08 ± 0.18
$NH_4^{\pm}$ (mg/L-N)	50.0 ± 4.9	42.4 ± 4.8	44.3 ± 4.1
NO_3^- (mg/L-N)	0.025 ± 0.003	0.032 ± 0.007	0.0030 ± 0.006
COD (mg/L)	168.3 ± 20.1	150.9 ± 14.0	167.9 ± 9.3
BOD$_5$ (mg/L)	88.7 ± 15.8	88.7 ± 9.3	86.4 ± 12.0
Total coliform (MPNs/100 mL)	1.9 * 10^6 ± 3.8 * 10^5	1.4 * 10^6 ± 4.0 * 10^5	1.3 * 10^6 ± 4.4 * 10^5
E. coli (MPNs/100 mL)	7.9 * 10^5 ± 3.0 * 10^5	7.3 * 10^5 ± 2.0 * 10^5	8.5 * 10^4 ± 2.9 * 10^4

Table 2. Percent removal of contaminants from RC constructed wetlands during the 0.5, 1, and 2 d HRT experiments.

	$RC_1$1			RC_2		
Parameter	0.5 d HRT	d HRT	2 d HRT	0.5 d HRT	1 d HRT	2 d HRT
TSS	88 ± 3%	90 ± 3%	93 ± 2%	92 ± 5%	94 ± 4%	95 ± 3%
Turbidity	72 ± 6%	88 ± 3%	91 ± 4%	93 ± 2%	96 ± 1%	98 ± 1%
PO_4^{3-}	40 ± 6%	69 ± 3%	84 ± 3%	71 ± 9%	97 ± 1%	92 ± 7%
$NH_4^{\pm}$	39 ± 8%	84 ± 2%	92 ± 3%	78 ± 8%	99 ± 0%	96 ± 3%
NO_3^-	35 ± 9%	37 ± 14%	88 ± 3%	30 ± 16%	54 ± 9%	35 ± 13%
COD	71 ± 6%	69 ± 4%	88 ± 3%	83 ± 11%	78 ± 3%	88 ± 4%
BOD_5	85 ± 2%	93 ± 2%	97 ± 1%	96 ± 2%	98 ± 1%	98 ± 1%
Total coliform (log removal)	0.74 ± 0.4	1.48 ± 0.26	2.90 ± 0.43	1.31 ± 0.20	2.17 ± 0.34	3.04 ± 0.40
E.coli (log removal)	0.79 ± 0.24	2.30 ± 0.29	3.05 ± 0.38	1.58 ± 0.31	3.58 ± 0.39	3.76 ± 0.39

Table 3. Percent removal of contaminants from CL constructed wetlands during the 0.5, 1, and 2 d HRT experiments.

	CL_1			CL_2		
Parameter	0.5 d HRT	1 d HRT	2 d HRT	0.5 d HRT	1 d HRT	2 d HRT
TSS	82 ± 3%	71 ± 5%	77 ± 5%	90 ± 2%	85 ± 4%	93 ± 2%
Turbidity	53 ± 8%	73 ± 8%	81 ± 7%	90 ± 2%	90 ± 2%	95 ± 2%
PO_4^{3-}	51 ± 4%	72 ± 2%	86 ± 2%	94 ± 2%	96 ± 1%	98 ± 1%
NH_4^+	70 ± 5%	89 ± 1%	93 ± 4%	97 ± 4%	99 ± 0%	96 ± 4%
NO_3^-	23 ± 14%	43 ± 12%	26 ± 17%	39 ± 10%	58 ± 8%	56 ± 10%
COD	61 ± 8%	45 ± 11%	69 ± 6%	78 ± 6%	69 ± 7%	87 ± 4%
BOD_5	80 ± 3%	79 ± 8%	88 ± 3%	95 ± 1%	95 ± 2%	98 ± 1%
Total coliform (log removal)	0.74 ± 0.14	1.54 ± 0.24	2.30 ± 0.27	1.05 ± 0.08	2.37 ± 0.31	2.66 ± 0.36
E. coli (log removal)	0.66 ± 0.23	2.57 ± 0.17	2.12 ± 0.56	1.30 ± 0.27	4.22 ± 0.44	3.31 ± 0.65

Table 4. Percent removal of contaminants from NP constructed wetlands during the 0.5, 1, and 2 d HRT experiments.

	NP_1			NP_2		
Parameter	0.5 d HRT	1 d HRT	2 d HRT	0.5 d HRT	1 d HRT	2 d HRT
TSS	91 ± 2%	91 ± 3%	84 ± 7%	94 ± 1%	94 ± 4%	93 ± 3%
Turbidity	74 ± 4%	93 ± 1%	89 ± 4%	91 ± 1%	95 ± 1%	95 ± 2%
PO_4^{3-}	9 ± 6%	14 ± 7%	32 ± 15%	18 ± 5%	24 ± 5%	57 ± 8%
NH_4^+	12 ± 8%	56 ± 4%	76 ± 6%	46 ± 10%	93 ± 1%	96 ± 3%
NO_3^-	42 ± 7%	24 ± 20%	51 ± 54%	45 ± 17%	227 ± 123%	141 ± 56%
COD	75 ± 5%	75 ± 5%	81 ± 4%	83 ± 4%	83 ± 3%	89 ± 3%
BOD_5	87 ± 4%	89 ± 5%	84 ± 6%	90 ± 1%	97 ± 1%	98 ± 1%
Total coliform (log removal)	0.80 ± 0.12	1.49 ± 0.26	2.28 ± 0.24	1.05 ± 0.13	2.44 ± 0.43	2.61 ± 0.21

performance of the CL constructed wetlands can be attributed to the root structure of *C. lacryma-jobi*. The roots of *C. lacryma-jobi* at maturity are relatively long (25–50 cm) and wide (∼3 mm) compared to the much shorter (∼3–9 cm) and thinner *R. corymbosa* roots (Figure 3) (Sahid and Hossain 1995; Jampeetong *et al.* 2013; Chen *et al.* 2015). Thus, it can be concluded that the larger and longer root system in the CL constructed wetlands created a preferential flow path for solids transport compared to the RC and NP constructed wetlands which have smaller to no root structures and thus limited preferential flow paths.

Orthophosphate removal

After the first pass under the 0.5-day HRT (Figure 4), CL_1 removed significantly more (<0.05) PO_4^{3-} (51 ± 4%) than RC_1 which removed significantly more PO_4^{3-} (40 ± 6%) than NP_1 (9 ± 6%). During the longer retention times, there were no statistically significant differences in PO_4^{3-} removal between CL_1 and RC_1 but both removed significantly more ($p < 0.05$) PO_4^{3-} than NP_1. After the first pass under a 1-day HRT, PO_4^{3-} removals of 72 ± 2, 69 ± 3, and 14 ± 7% were observed for CL_1, RC_1, and NP_1, respectively. After the first pass under a 2-day HRT, PO_4^{3-} removals improved to 86 ± 2, 84 ± 3, and 32 ± 15% for CL_1, RC_1, and NP_1, respectively.

After the second pass under all three HRTs, PO_4^{3-} removal significantly increased ($p < 0.05$) from the first pass for the CL, RC, and NP constructed wetlands with removal percentages ranging from 94 ± 2 to 98 ± 1, 71 ± 9 to 97 ± 1, and 18 ± 5 to 57 ± 8%, respectively. The effectiveness of the various constructed wetlands during the second pass followed a similar trend as was observed during the first pass. Primarily, under the 0.5-day HRT, the CL_2 constructed wetland was more effective at remove PO_4^{3-} than the RC_2 constructed wetland but they both became equally effective at removing PO_4^{3-} as the HRTs increased to 1 and 2 days. Under all HRTs tested, both CL_2 and RC_2 were more effective at removing PO_4^{3-} than NP_2.

The PO_4^{3-} removals observed in this study fall within the 50–83% (60.1% average) removals that have been previously reported for tropical/sub-tropical vertical flow constructed wetlands treating municipal wastewater (Zhang

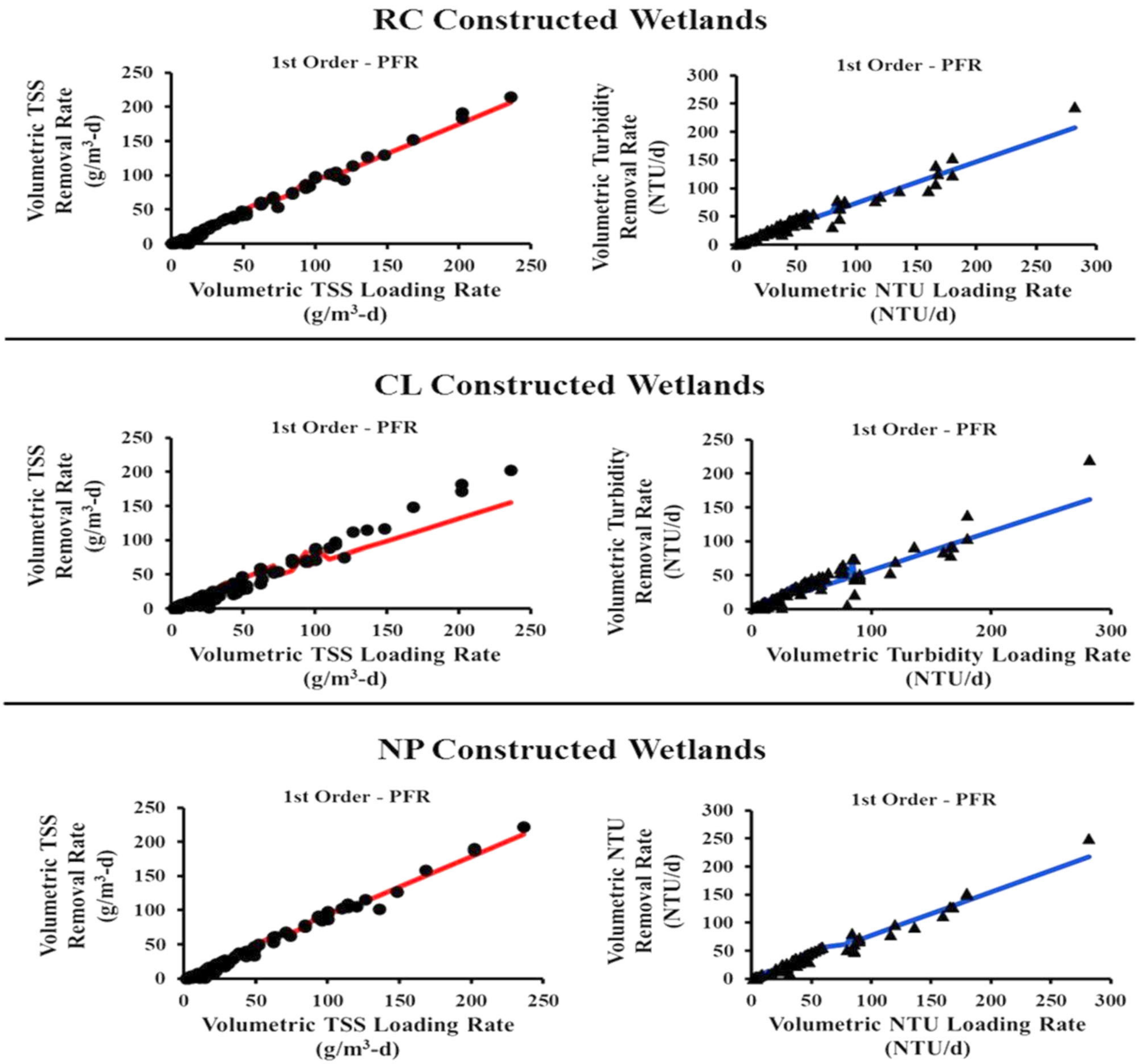

Figure 2. The 1st order PFR model fit of volumetric removal rates *vs.* volumetric loading rates for TSS (●) and turbidity (▲) in RC, CL, and NP constructed wetlands.

Table 5. Kinetic rate constants for RC constructed wetlands.

Volumetric removal rates (RC)									
	1st Order—PFR				Stover-Kincannon				
Contaminant	K (d^{-1})	R^2 (0–1)	RRMSE (0 to INF)	ME (−INF to 1)	U_{MAX} (g/m^3-d)	K_B (g/m^3-d)	R^2 (0–1)	RRMSE (0 to INF)	ME (−INF to 1)
NH_4^+-N	2.4	0.70	0.58	0.50	79.4	65.8	0.74	0.41	0.74
PO_4^{3-}-P	1.1	0.72	0.36	0.71	0.27	0.22	0.70	0.34	0.75
COD	1.5	0.90	0.47	0.82					
BOD_5	3.5	0.99	0.07	0.99					
TSS	4.2	0.99	0.16	0.99					
Turbidity (NTU/d)	2.7	0.97	0.21	0.97					
E. coli (MPN/100 mL-d)	4.2	0.98	0.25	0.98					
Total coliform (MPN/100 mL-d)	3.8	0.99	0.16	0.99					

et al. 2015). The superior removal of PO_4^{3-} from CL and RC constructed wetlands compared to NP constructed wetlands under all conditions tested demonstrates the importance of plant in the removal of PO_4^{3-} from wastewater. The complexity of PO_4^{3-} removal in constructed wetlands through multiple pathways have been reported, including adsorption and precipitation. The presence of plants enhanced PO_4^{3-} removal has also been confirmed (Xu and Li 2007; Stefanakis and Tsihrintzis 2012; Zhang *et al.* 2015; Papaevangelou *et al.* 2016). The Stover-Kincannon model most accurately predicted PO_4^{3-} removal in RC and CL constructed wetlands (Tables 5

Table 6. Kinetic rate constants for CL constructed wetlands.

	Volumetric removal rates (CL)								
	1st Order—PFR				Stover-Kincannon				
Contaminant	k (d^{-1})	R^2 (0–1)	RRMSE (0 to INF)	ME (−INF to 1)	U_{MAX} (g/m^3-d)	K_B (g/m^3-d)	R^2 (0–1)	RRMSE (0 to INF)	ME (−INF to 1)
NH_4^+-N	2.5	0.96	0.17	0.96	261.6	256.3	0.97	0.16	0.99
PO_4^{3-}-P	1.5	0.82	0.31	0.81	0.47	0.42	0.88	0.23	0.90
COD	0.8	0.69	0.69	0.54					
BOD_5	2.3	0.94	0.18	0.93					
TSS	2.2	0.94	0.38	0.91					
Turbidity (NTU/d)	1.7	0.90	0.34	0.89					
E. coli (MPN/100 mL-d)	3.8	0.95	0.39	0.95					
Total coliform (MPN/100 mL-d)	3.8	0.98	0.19	0.98					

Table 7. Kinetic rate constants for NP constructed wetlands.

	Volumetric removal rates (NP)								
	1st Order—PFR				Stover-Kincannon				
Contaminant	k (d^{-1})	R^2 (0–1)	RRMSE (0 to INF)	ME (−INF to 1)	U_{MAX} (g/m^3-d)	K_B (g/m^3-d)	R^2 (0–1)	RRMSE (0 to INF)	ME (−INF to 1)
NH_4^+-N	0.8	0.52	0.44	0.47	40.1	26.7	0.56	0.41	0.88
PO_4^{3-}-P	0.2	0.23	0.99	0.19	0.07	0.22	0.11	1.03	0.11
COD	1.9	0.95	0.37	0.90					
BOD_5	3.5	0.99	0.10	0.98					
TSS	4.5	0.99	0.16	0.99					
Turbidity (NTU/d)	3	0.99	0.17	0.98					
E. coli (MPN/100 mL-d)	4.1	0.96	0.27	0.98					
Total coliform (MPN/100 mL-d)	3.8	0.99	0.18	0.98					

Figure 3. The root structure of *R. corymbosa* (left) and *C. lacryma-jobi* (right) three months after germination.

and 6). The predicted maximum PO_4^{3-} volumetric removal rate (UMAX) for CL constructed wetlands was nearly twice that for RC constructed wetlands at 0.47 and 0.27 g/m^3-d, respectively. That was expected as *C. lacryma-jobi* has a much larger average above-ground and below-ground biomass than *R. corymbosa* (Jampeetong *et al.* 2013; Chen *et al.* 2015; Taha *et al.* 2015). Also, when compared on a per gram dry weight basis, the phosphorus content of *C. lacryma-jobi* roots and leaves are 4–7 and 8–15 mg/g dw, respectively (Xiao-Yin *et al.* 2010; Jampeetong *et al.* 2013). This is considerably higher than the phosphorus content of *R. corymbosa* roots and leaves at roughly 2.5 mg/g dw each (Greenway and Woolley 1999). Thus, it can be concluded that the CL constructed wetlands had faster PO_4^{3-} uptake rates due to the larger size and phosphorus requirements of *C. lacryma-jobi* compared to *R. corymbosa*.

Unlike the RC and CL constructed wetlands, PO_4^{3-} removal in NP constructed wetlands were most accurately represented by 1st order removal (Table 7). This, in combination with the significantly lower PO_4^{3-} removal percentages compared to CL and RC constructed wetlands, suggests that PO_4^{3-} removal in NP constructed wetlands is primarily due to abiotic processes. These removal processes include sorption to the granular media, soil, and trapped sediment, which have previously been reported as important PO_4^{3-} removal mechanisms in constructed wetlands (Zhang *et al.* 2015; Zhu *et al.* 2017).

Nitrogen removal

The CL_1 removed significantly more NH_4^+ (70 ± 5 to 89 ± 1%) than RC_1 which removed significantly more NH_4^+ (39 ± 8 to 84 ± 2%) than NP_1 (12 ± 8 to 56 ± 4%) after the

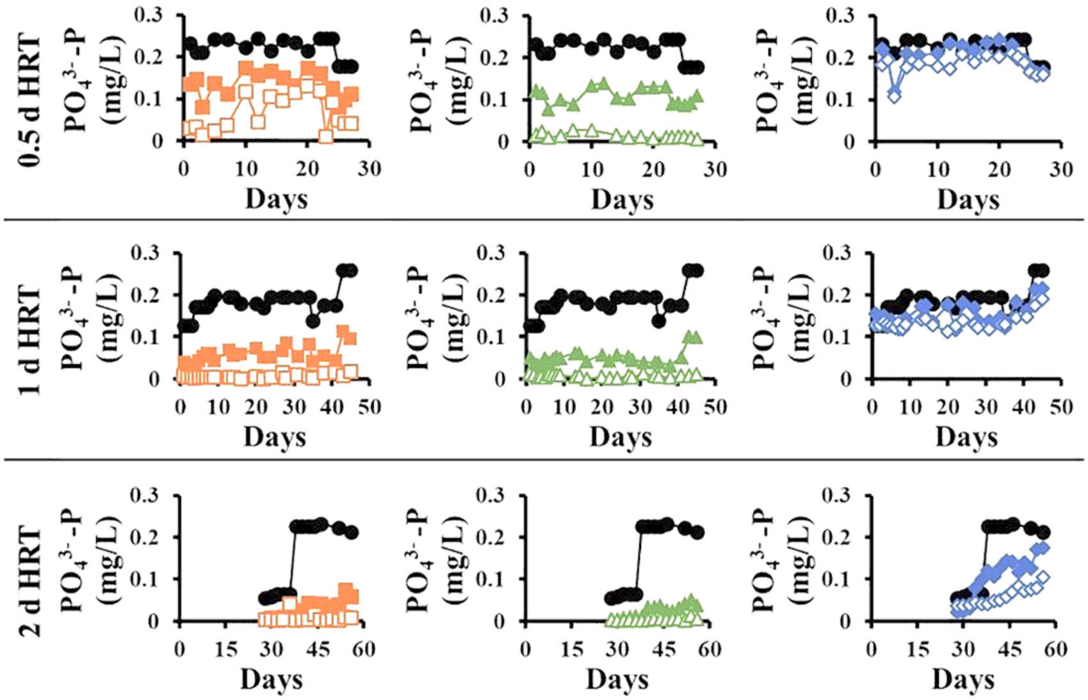

Figure 4. The PO_4^{3-}-P removal from the influent (●) and effluent PO_4^{3-}-P removal from RC_1 (■), RC_2 (□), CL_1 (▲), CL_2 (△), NP_1 (◆), and NP_2 (◇).

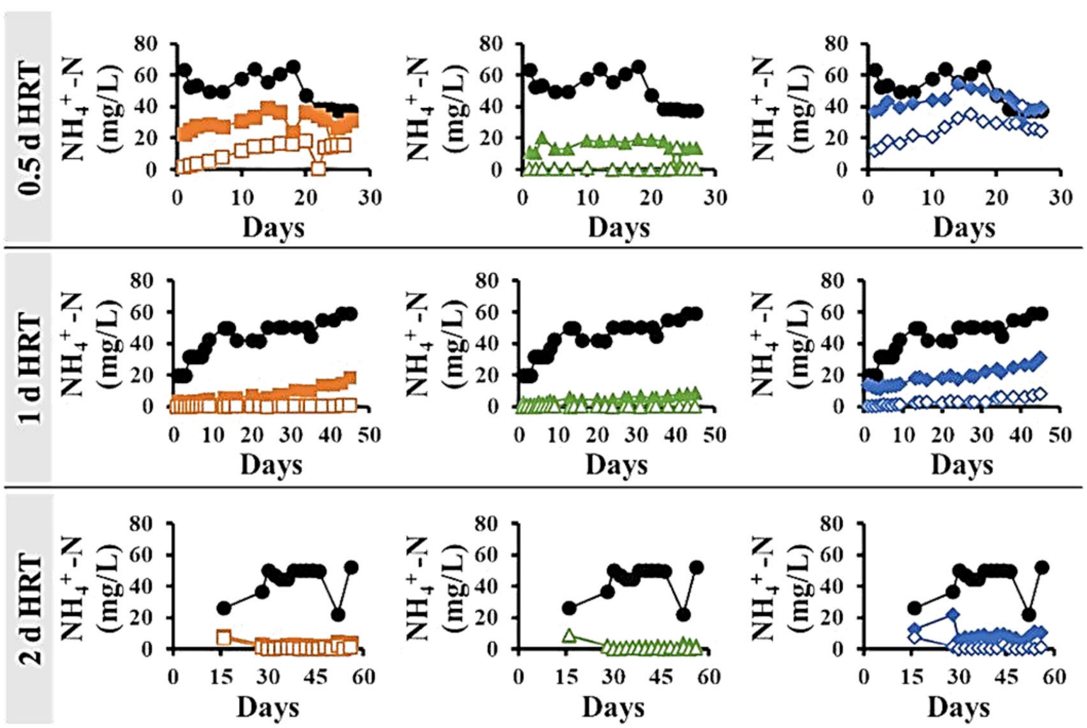

Figure 5. NH_4^+-N removal; the influent NH_4^+-N (●) and effluent NH_4^+-N from RC_1 (■), RC_2 (□), CL_1 (▲), CL_2 (△), NP_1 (◆), and NP_2 (◇).

first pass under the 0.5 day and 1 day HRT conditions (Figure 5). For the 2-day HRT condition, CL_1 and RC_1 were equally effective in NH_4^+ removal (93 ± 4 and $92 \pm 3\%$, respectively) and both were significantly more effective than NP_1 ($76 \pm 6\%$). Similar results were observed after the second pass under 0.5-day HRT conditions with CL_2 removing more NH_4^+ ($97 \pm 4\%$) than RC_2 which removed NH_4^+ ($78 \pm 8\%$) than NP_2 ($46 \pm 10\%$). At a 1-day HRT, CL_2, and RC_2 removed statistically similar ($p > 0.05$) amounts of NH_4^+ ($99 \pm 0\%$), and both performed statistically better than NP_2 ($93 \pm 1\%$). At a 2-day HRT, there was no statistical difference in NH_4^+ removal between CL_2, RC_2, and NP_2 (ranging from 93 ± 4 to $96 \pm 3\%$).

In the case of NO_3^-, under all three HRT conditions, NO_3^- removal was statistically similar between CL_1 and RC_1 with removals ranging from 23 ± 14 up to $43 \pm 12\%$ (Figure

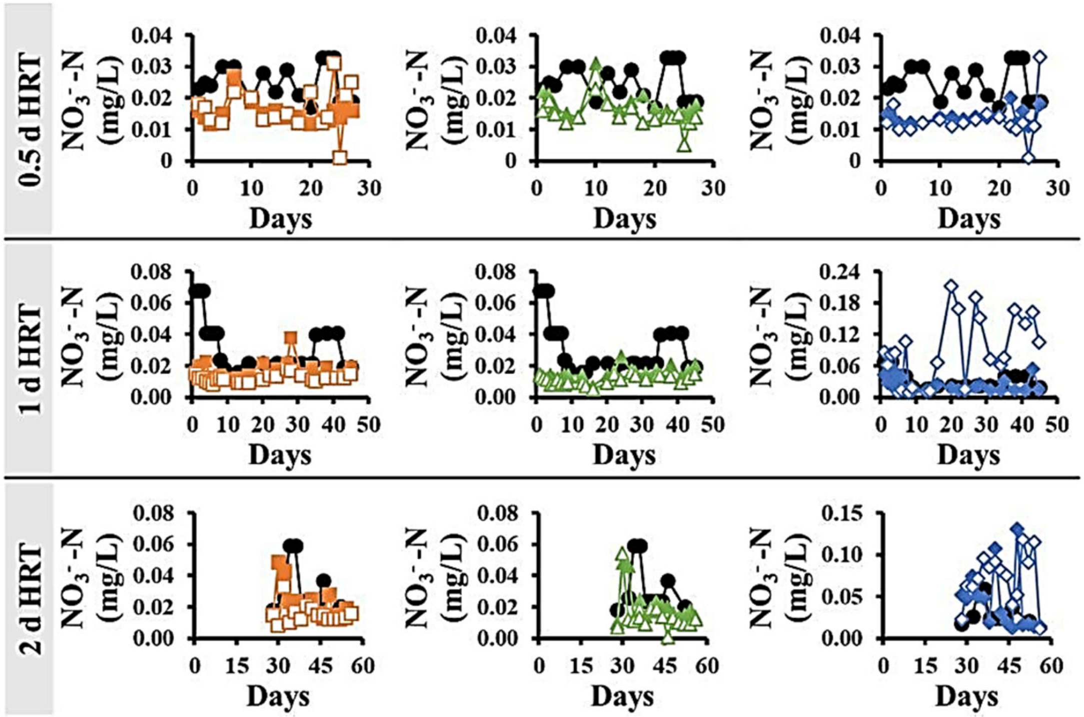

Figure 6. NO_3^--N removal; the influent NO_3^--N (●) and effluent NO_3^--N from RC_1 (■), RC_2 (□), CL_2 (▲), CL_2 (△), NP_1 (◆), and NP_2 (◇).

6). NP_1 behaved statistically similar to CL_1 and RC_1 under the 0.5 and 1-day HRT conditions with NO_3^- removals of 42 ± 7 and $24 \pm 20\%$, respectively. Unlike the 0.5 and 1-day HRTs, under the 2-day HRT, NP_1 performed significantly worse than CL_1 and RC_1 with the effluent NO_3^- concentrations increased by an average of $51 \pm 54\%$. Similar to the first pass, during the second pass under all three HRT conditions, NO_3^- removal was statistically similar between CL_2 and RC_2 with removals ranging from 30 ± 16 up to $58 \pm 8\%$. Under the 0.5-day HRT condition, NP_2 performed similarly to CL_2 and RC_2 with a NO_3^- removal of $45 \pm 17\%$. However, with a longer HRT of 1 and 2 days, the NO_3^- concentration in the effluent of NP_2 increased significantly up to 227 ± 123 and $141 \pm 56\%$, respectively. It should be noted that although the observed increases in NO_3^- concentrations are statistically significant, they represent <1%, on a mass basis, of the NH_4^+ being removed in these systems and it does not significantly decrease the observed total nitrogen removal percentages in these systems. The typical NH_4^+ removals that have been previously reported for tropical/sub-tropical vertical flow constructed wetlands treating municipal wastewater range from 46 to 75% (66% average) (Zhang *et al.* 2015). The NH_4^+ removals observed in this study bracket that range and varied considerably by plant type. The NH_4^+ volumetric removal rates are best described by Stover-Kincannon kinetics with predicted maximum NH_4^+ volumetric removal rates (UMAX) of 79.4, 261.6, and 40.1 g/m^3-d for RC, CL, and NP constructed wetlands, respectively (Tables 1–3, Figure 6). The superior removal of NH_4^+ and NO_3^- from CL and RC constructed wetlands compared to NP constructed wetlands under all conditions tested demonstrates the importance of plant in the removal of NH_4^+ and NO_3^- from wastewater, as has been noted previously in the literature (Brix *et al.* 2002; Keffala and Ghrabi 2005; Stefanakis and Tsihrintzis 2012; Ge *et al.* 2015; Zhang *et al.* 2015; Papaevangelou *et al.* 2016).

Nevertheless, the NP constructed wetlands also removed large quantities of NH_4^+ indicating that removal mechanisms other than plant uptake also play an important role in NH_4^+ removal. Part of the importance of plants for NH_4^+ removal is that they provide habitat for autotrophic nitrifying bacteria in their rhizosphere, which oxidize NH_3 to NO_3^-, the preferred nitrogen source for many plant species, including *C. lacryma-jobi* (Jampeetong *et al.* 2013). The planted constructed wetlands experienced significant decreases ($p < 0.05$) in pH with greater pH drops observed with increased HRTs under the 2-day HRT, the pH dropped from 7.1 ± 0.2 down to 6.2 ± 0.1 for both CL_1 and RC_1.

These drops in pH, in combination with the abundance of dissolved oxygen (DO) can be attributed to nitrification overwhelming the natural alkalinity of the wastewater (Tchobanoglous *et al.* 2003). Additionally, the nitrogen requirement of the plants themselves is a major component of the improved nitrogen removal in the CL and RC constructed wetlands over the NP constructed wetlands. The greater nitrogen removals by CL constructed wetlands compared to RC constructed wetlands can be attributed to the higher nitrogen requirements of *C. lacryma-jobi* compared to *R. corymbosa*. Similar to phosphorus, the nitrogen content of *C. lacryma-jobi* roots and leaves (25–50 and 35–60 mg/g dw, respectively) are considerably higher than that for *R. corymbosa* roots and leaves (roughly 14 and 17 mg/g dw, respectively) (Greenway and Woolley 1999; Xiao-Yin *et al.* 2010; Jampeetong *et al.* 2013).

Unlike the planted constructed wetlands, there was no significant drop in pH under any condition in the NP constructed wetlands even though DO levels were high and favorable for nitrification to occur. This suggests that nitrification was not strong without the presence of plants. However, there was a slight increase in NO_3^- in the effluent of the NP constructed wetlands, indicated that some minor nitrification was occurring as shown in Figure 6. The build-up of NO_3^- in the presence of high DO levels (2–7 mg/L) and relatively low BOD_5 concentrations (1–25 mg/L) suggests that heterotrophic denitrification is not significant in these systems. The lack of significant nitrification and heterotrophic denitrification activity in the NP constructed wetlands suggests that their removal of NH_4^+ is primarily due to adsorption to the gravel media (Vymazal 2011b; Saeed and Sun 2011).

The maximum ammonium adsorption capacity for gravel has been reported in the range of 400–800 mg/kg (Zhu *et al.* 2012). The constructed wetlands used in this study contained 86 kg of gravel and could be expected to remove 35–70 g of NH_4^+ before reaching adsorptive capacity. The NP_1 constructed wetland removed a total of 50 g of NH_4^+, which is within the expected range of the gravel beds. There is also a time dependence on NH_4^+ removal in NP constructed wetlands with longer HRTs resulting in higher removals. A previous study by Zhu *et al.* (2012) showed that NH_4^+ adsorption did not stabilize on gravel until after 12–24 h of exposure. Thus, the poor removal in the 0.5 d HRT NP constructed wetlands may be due to kinetic rate limitations.

Organic carbon removal

The RC_1 and NP_1 showed similar performance by achieving 69 ± 4 to 81 ± 4% COD removal with CL_1 performing significantly worse ($p < 0.05$) by achieving only 45 ± 11 to 69 ± 6% COD removal after the first pass, under all three HRT conditions as shown in Figure 7 and Tables 2–4. After passing through the second constructed wetland under all conditions tested, the COD removal percentages increased to the range of 69 ± 7 to 89 ± 3% and there was no significant difference ($p > 0.05$) between the two plant conditions and no plant control. However, there was one exception. Under a 1-day HRT, the CL_2 effluent contained 42.2 ± 6.6 mg/L COD, which was significantly higher ($p < 0.05$) than the RC_2 effluent (31.8 ± 4.7 mg/L) which is significantly higher ($p < 0.05$) the NP_2 effluent (23.2 ± 3.1).

The BOD_5 removal was more variable than COD removal as shown in Figure 8. After the first pass, under the 0.5-day HRT, the RC_1, CL_1, and NP_1 demonstrated statistically similar ($p > 0.05$) performance by achieving 80 ± 3 to 87 ± 4% BOD5 removal. Under the 1-day HRT, RC and NP performed significantly better ($p < 0.05$) than CL_1 at 93 ± 2, 89 ± 5, and 79 ± 8% BOD_5 removal, respectively. Under the 2-day HRT, RC_1 performed significantly better ($p < 0.05$) than both NP_1 and CL_1 at 97 ± 1, 84 ± 6, and 88 ± 3% BOD_5 removal, respectively. The BOD_5 concentrations significantly decreased ($p < 0.05$) after passing through the second constructed wetland under all conditions tested, with BOD_5 removal percentages increased to the range of 90 ± 1 to 98 ± 1%. In general, after the second pass, there was no significant difference ($p > 0.05$) between the two plant conditions and the no plant control under each HRT condition. The COD and BOD_5 removals observed in this study fall within the removal ranges [63–91% (64.1% average) and 81–96% (87.6% average), respectively] that have been previously reported for tropical/sub-tropical vertical flow constructed wetlands treating municipal wastewater (Zhang *et al.* 2015).

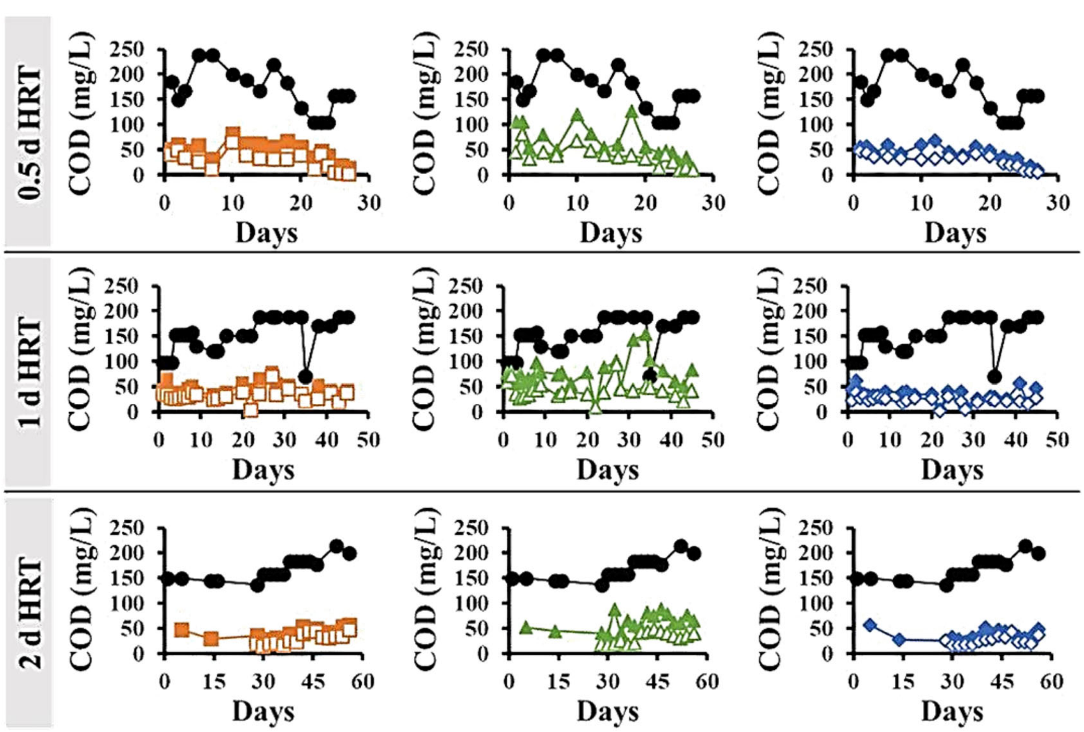

Figure 7. COD removal; the influent COD (●) and effluent COD from RC_1 (■), RC_2 (□), CL_1 (▲), CL_2 (△), NP_1 (◆), and NP_2 (◇).

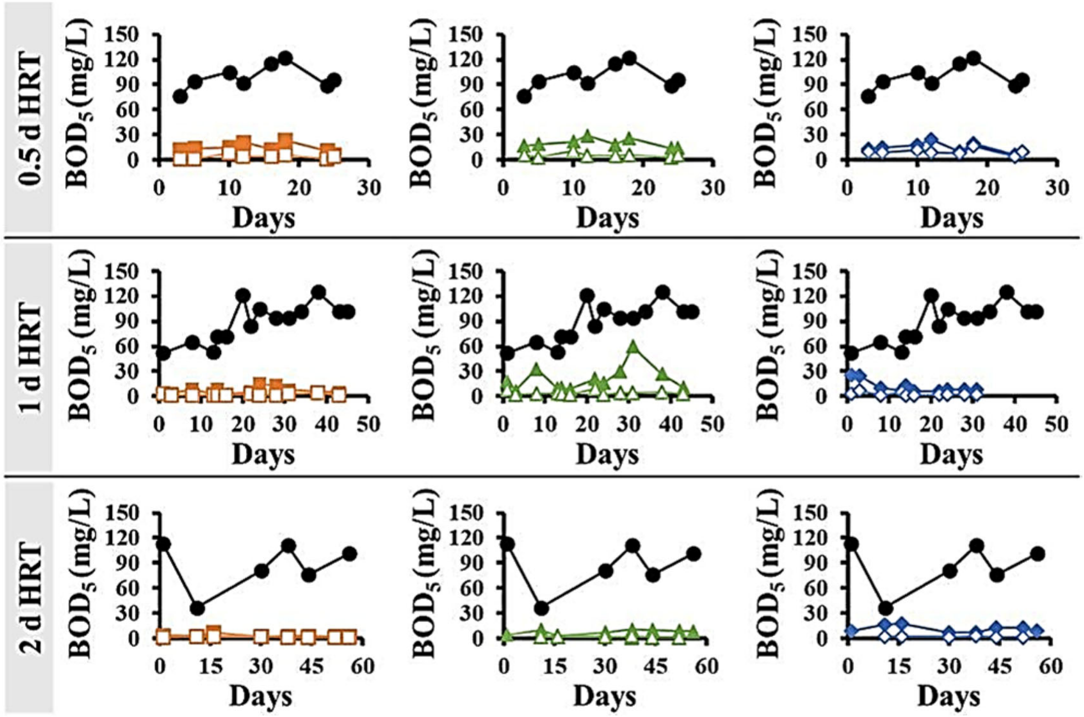

Figure 8. BOD_5 removal; the influent BOD_5 (●) and effluent BOD_5 from RC_1 (■), RC_2 (□), CL_1 (▲), CL_2 (△), NP_1 (◆), and NP_2 (◊).

The COD and BOD_5 removal in constructed wetlands are complicated processes involving numerous mechanisms including biological degradation, adsorption, and filtration (Akratos and Tsihrintzis 2007; Prochaska *et al.* 2007). Several studies have documented that COD and BOD_5 removal in vertical flow constructed wetlands can be enhanced by the presence of plants due to their root structures providing additional surface area and oxygen to microbes, increasing their abundance and activity (Chang *et al.* 2012; Papaevangelou *et al.* 2016). However, the results of this study appear to be more nuanced.

The COD and BOD_5 volumetric removal rates in each of the constructed wetlands can be accurately characterized using 1st order kinetics as shown in Figure 9. For the case of COD removal, the NP constructed wetlands had the greatest 1st order rate constant (1.9 d^{-1}), followed by the RC constructed wetlands (1.5 d^{-1}), and then the CL constructed wetlands (0.8 d^{-1}). Similar trends are observed for BOD_5 with NP and RC constructed wetlands having higher 1st order rate constants (3.5 d^{-1}, each) than the CL constructed wetlands (2.3 d^{-1}). The slower COD and BOD_5 removal rates of the CL constructed wetlands may be due to the relatively long and wide *C. lacryma-jobi* roots compared to the smaller fibrous *R. corymbosa* roots in the RC constructed wetlands, or the lack of roots in the NP constructed wetlands (Sahid and Hossain 1995; Jampeetong *et al.* 2013; Chen *et al.* 2015). This suggests that filtration is an important mechanism for COD and BOD_5 removal and that the larger and wider *C. lacryma-jobi* roots created preferential flow paths for COD and BOD_5 to travel through the system. This is a similar phenomenon to what was observed in regard to TSS and turbidity removal with the CL constructed wetlands often performing more poorly than the RC and NP constructed wetlands.

Coliform removal

After the first pass, the RC_1, CL_1, and NP_1 demonstrated similar total coliform log removals ($p > 0.05$) of 0.74 ± 0.14 to 0.80 ± 0.12, 1.48 ± 0.26 to 1.54 ± 0.24 and 2.28 ± 0.24 to 2.90 ± 0.43 for the 0.5, 1, and 2-day HRTs, respectively as shown in Figure 10 and Tables 2–4. The total coliform concentrations significantly decreased ($p < 0.05$) after passing through the second constructed wetland under all conditions tested, with RC_2, CL_2, and NP_2 demonstrating similar total coliform removals ($p > 0.05$). The second pass achieved total coliform log removals of 1.05 ± 0.13 to 1.31 ± 0.20, 2.17 ± 0.34 to 2.44 ± 0.43, and 2.61 ± 0.21 to 3.04 ± 0.40 for the 0.5, 1, and 2-day HRTs, respectively.

Similar to total coliform, after the first pass, the RC_1, CL_1, and NP_1 demonstrated similar *E. coli* log removals ($p > 0.05$) of 0.74 ± 0.14 to 0.80 ± 0.12, 1.48 ± 0.26 to 1.54 ± 0.24, and 2.28 ± 0.24 to 2.90 ± 0.43 for the 0.5, 1, and 2-day HRTs, respectively (Figure 11 and Tables 2–4). The *E. coli* concentrations significantly decreased ($p < 0.05$) after passing through the second constructed wetland under all conditions tested, with RC_2, CL_2, and NP_2 demonstrating similar log removals ($p > 0.05$) of 0.66 ± 0.23 to 0.79 ± 0.24, 2.30 ± 0.29 to 2.57 ± 0.17, and 2.12 ± 0.56 to 3.05 ± 0.38 for the 0.5, 1, and 2-day HRTs, respectively. For all constructed wetlands, there was a time dependent removal of both total coliform and *E. coli* with higher removals observed at longer HRTs, which has been observed previously in the literature (Tanner *et al.* 1995; Green *et al.* 1997; García *et al.* 2003;

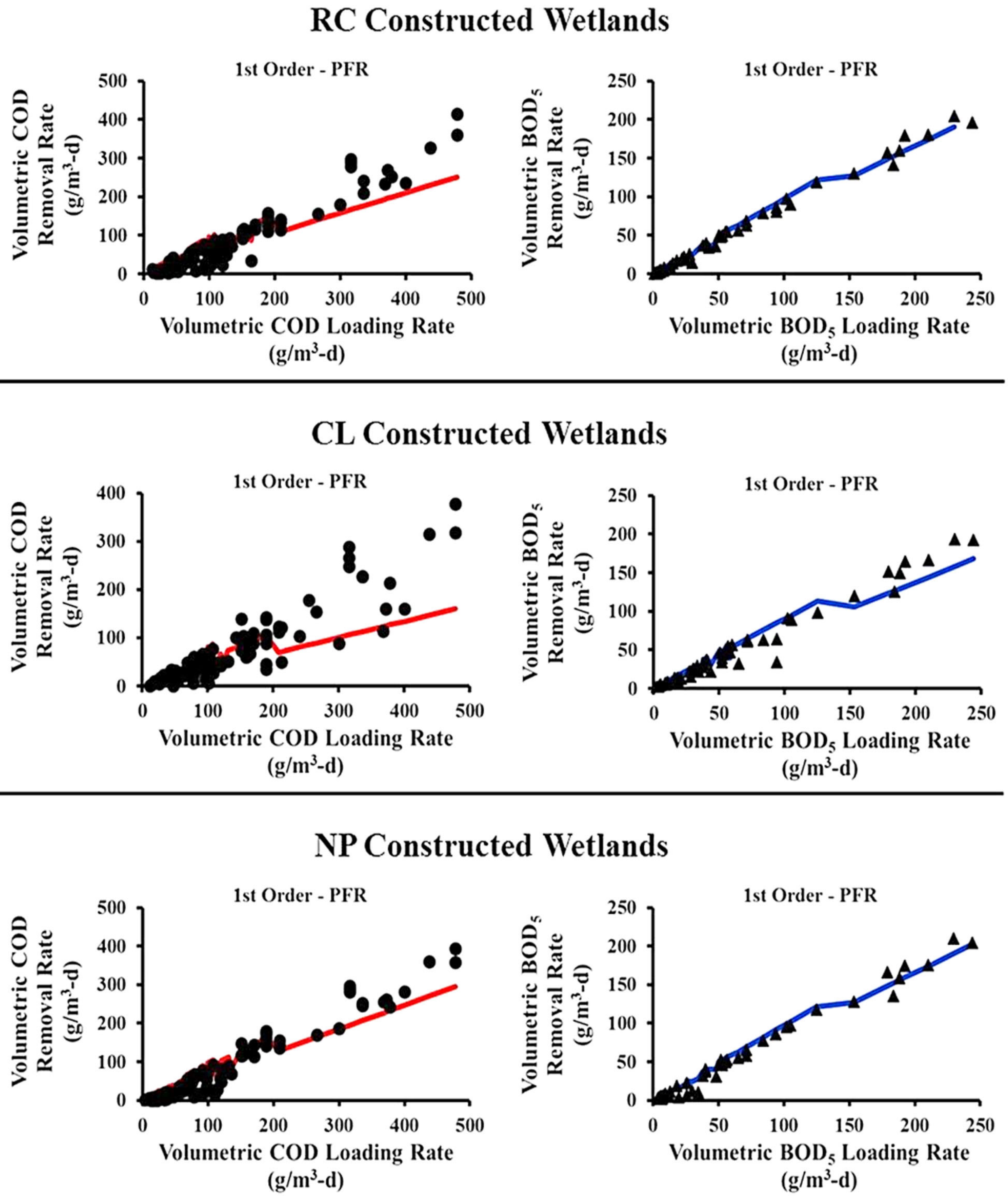

Figure 9. The 1st order PFR model fit of volumetric removal rates *vs.* volumetric loading rates for COD (●) and BOD_5 (▲).

Vymazal 2005). A wide variety of mechanisms have been proposed for coliform removal in constructed wetlands including physical removal (*e.g.*, filtration, sedimentation, and adsorption), toxicity (*e.g.*, biocides produced by plant roots and exposure to UV light), exposure to unfavorable aquatic conditions (*e.g.*, dissolved oxygen, pH, redox conditions) and endogenous decay (*e.g.*, temperature-related effects and predation) (MacIntyre *et al.* 2006; Weber and Legge 2008; Wu *et al.* 2016). While the CL constructed wetlands have consistently demonstrated inferior filtration capacity compared to the RC and NP constructed wetlands due to the *C. lacryma-jobi* root structure, they performed equally well in the removal of total coliforms and *E. coli* (Figures 10 and 11). This suggests that physical removal was not a dominant removal mechanism in these constructed wetlands. This is in agreement with previous studies that also determined that sedimentation and filtration was not major coliform removal mechanism in constructed wetlands, particularly those with gravel beds (Wand *et al.* 2007; Boutilier *et al.* 2009). The presence of plants has been

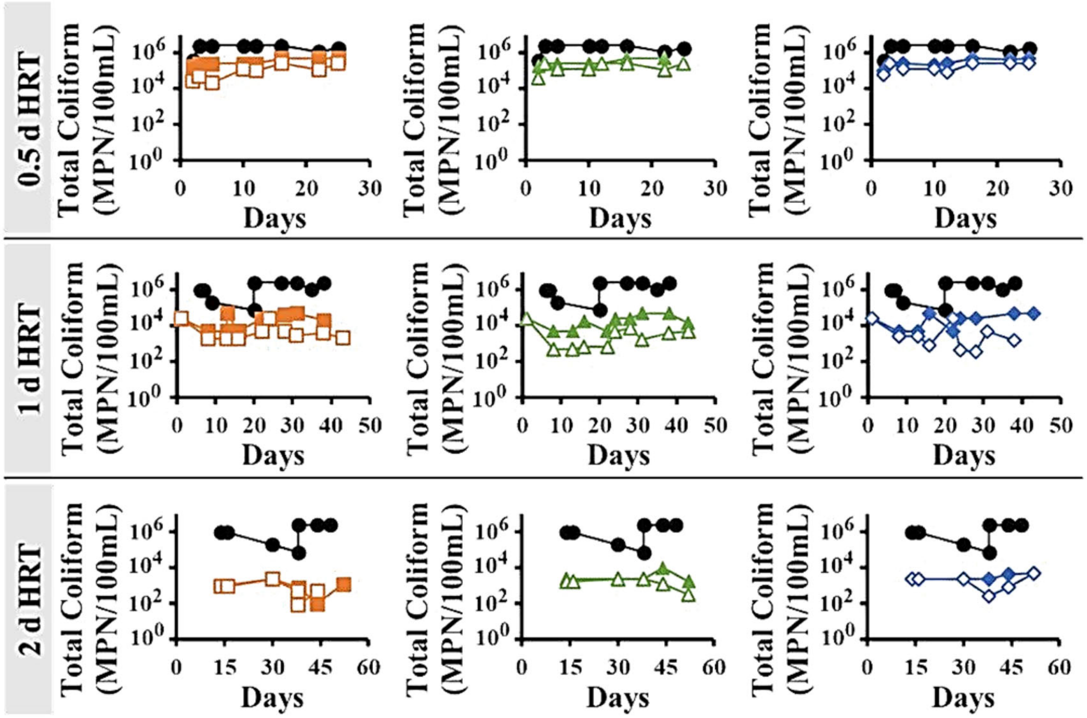

Figure 10. Total coliform removal; the influent total coliform (●) and effluent total coliform from RC_1 (■), RC_2 (□), CL_1 (▲), CL_2 (△), NP_1 (◆), and NP_2 (◊).

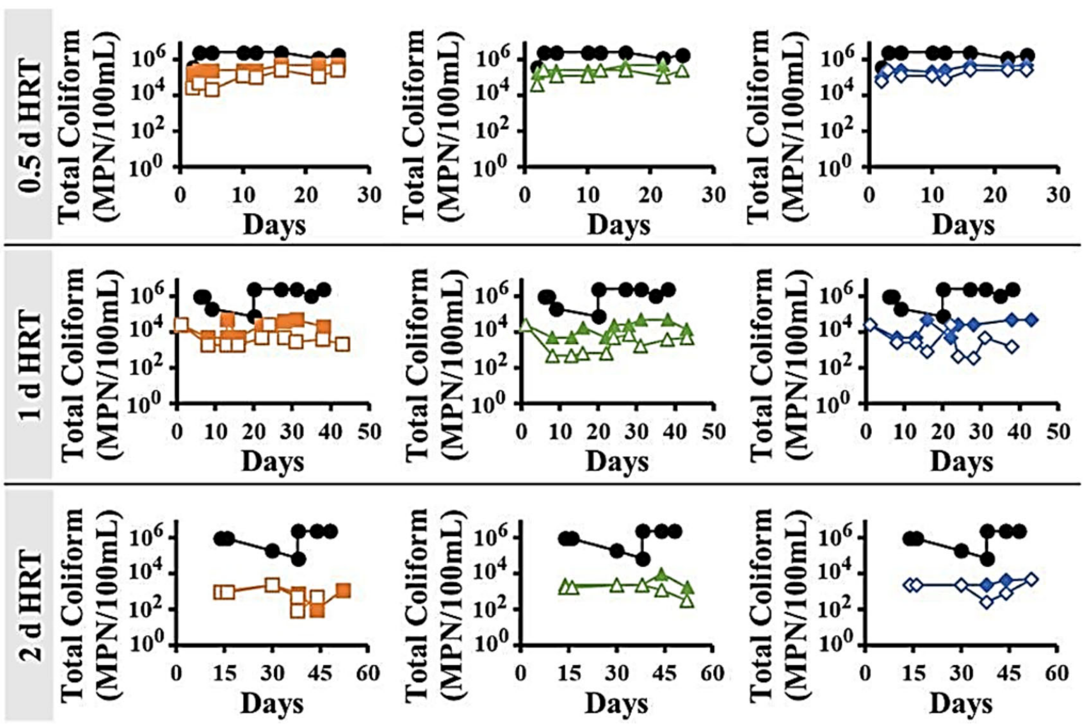

Figure 11. *E. coli* removal; the influent *E. coli* (●) and effluent *E. coli* from RC_1 (■), RC_2 (□), CL_1 (▲), CL_2 (△), NP_1 (◆), and NP_2 (◊).

suggested to enhance coliform removal through a variety of mechanisms including filtration and adsorption to the root systems and the release of biocides through the roots (Morsy *et al.* 2007; Weber and Legge 2008).

However, the equivalent performance between the non-planted and planted constructed wetland observed in this study can rule out plant biocides and enhanced root filtration/adsorption as major removal mechanisms. Several other studies have also reported that the presence of plants did not enhance coliform removal and in some cases, increased the survivability of coliforms (Karathanasis *et al.* 2003; Keffala and Ghrabi 2005; Wand *et al.* 2007; Wu *et al.* 2016). Pathogen removal by UV light exposure has been proposed to be a significant coliform removal mechanism in constructed wetlands, particularly those with the surface flow and/or standing water built into the design (MacIntyre *et al.*

2006; Weber and Legge 2008; Jenkins *et al.* 2011). However, in this study, the effect of UV light is assumed to be negligible due to the shading provided by the *C. lacryma-jobi* and plants in addition to the wastewater rapidly percolating to the subsurface of the constructed wetlands during each loading. This leaves endogenous decay and predation as possible mechanisms for total coliform and E. coli removal. Coliform removal *via* predation in constructed wetlands has widely been described using 1st order kinetics (Easton *et al.* 2005; Selvakumar *et al.* 2007; Struck *et al.* 2007; Boutilier *et al.* 2009; Chan *et al.* 2015). The *E. coli* and total coliform volumetric removal rates in this study follow 1st order decay with 1st order rate constants of 3.8 and 3.8–4.2 d^{-1} for total coliform and *E. coli*, respectively. These decay rates are similar to those previously reported for total coliform and *E. coli* removal in constructed wetlands (Struck *et al.* 2007; Chan *et al.* 2015). This suggests that predation may play an important role in E. coli and total coliform removal. Numerous studies have demonstrated predation as the primary removal mechanism for total coliform and *E. coli* in constructed wetlands (Decamp *et al.* 1999; Wand *et al.* 2007; Weber and Legge 2008; Jenkins *et al.* 2011; García *et al.* 2013). This is particularly true in tropical vertical flow constructed wetlands, such as in this study, that provide warm high oxygen environments that allow for protozoa to thrive (MacIntyre *et al.* 2006; Winward *et al.* 2008a, 2008b; Headley *et al.* 2013). The effluent dissolved oxygen concentrations in this study ranged from 2 ± 0.1 to 5.6 ± 0.4 mg/L and the effluent temperatures ranged from 22.5 ± 0.4 to 23.6 ± 0.3 °C.

Conclusions

The results of this study show that both *C. lacryma-jobi* and *R. corymbosa* planted vertical flow constructed wetlands are suitable for the treatment of municipal primary sedimentation effluent, although each plant type has different strengths and weaknesses in contaminant removal. Even though each type of constructed wetland was tested in isolation in this study, the results suggest that a combination of *C. lacryma-jobi* and *R. corymbosa* vertical flow constructed wetlands in series would provide an optimal design. This system could start with *R. corymbose* planted constructed wetlands which demonstrated maximum TSS, turbidity, COD, and BOD_5 removal, due to their shorter more fibrous roots, which minimizes preferential flow paths in the system. These constructed wetlands can be followed by *C. lacryma-jobi* vegetated constructed wetlands which demonstrated maximum nutrient removal due to their large biomass and high nitrogen and phosphorus content. To maximize pathogen removal, which is independent of plant type, the total HRT should be a minimum of 2 days. However, it should be noted that further disinfection may still be necessary to meet effluent requirements.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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